(13) $\lim_{x \to 1} \sin \pi x$

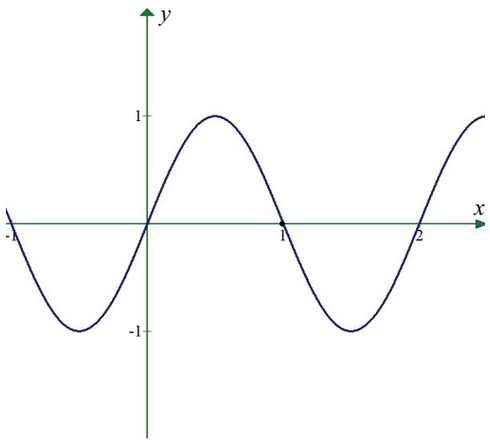


Fig. 9.19

(14) $\lim_{x \to 0} \sec x$

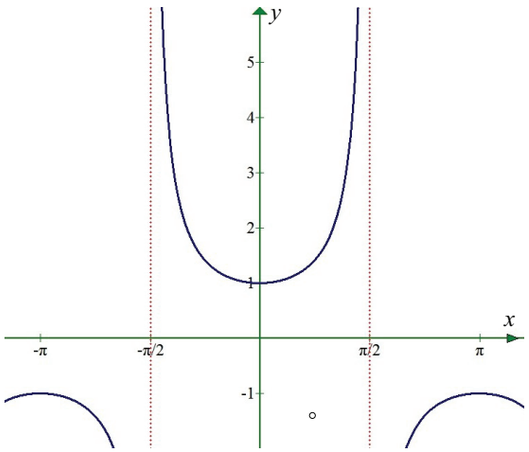


Fig. 9.20

(15) $\lim_{x \to \frac{\pi}{2}} \tan x$

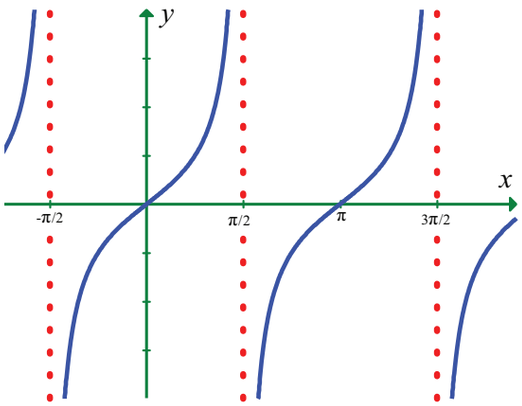


Fig. 9.21

Sketch the graph of f, then identify the values of x_0 for which $\lim_{x \to x_0} f(x)$ exists.

(16) $f(x) = \begin{cases} x^2, & x \le 2 \\ 8 - 2x, & 2 < x < 4 \\ 4, & x \ge 4 \end{cases}$.

(17) $f(x) = \begin{cases} \sin x, & x < 0 \\ 1 - \cos x, & 0 \le x \le \pi \\ \cos x, & x > \pi \end{cases}$

(18) Sketch the graph of a function f that satisfies the given values :

(i) $f(0)$ is undefined

$\lim_{x \to 0} f(x) = 4$

$f(2) = 6$

$\lim_{x \to 2} f(x) = 3$

(ii) $f(-2) = 0$

$f(2) = 0$

$\lim_{x \to -2} f(x) = 0$

$\lim_{x \to 2} f(x)$ does not exist.

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